67250 IS MILIEUX

Phase B: Web Design, Development & Implementation
Increment 3: Foundations of JavaScript

Overview

This increment focuses on integrating JavaScript into your museum website to introduce fundamental programming concepts and practice DOM manipulation.

By the end of this increment, you will have:

- A personalized, time-based greeting on the homepage.
- The current year automatically updated in the footer.

By the end of this increment, you will be able to:

- Link JavaScript to your existing HTML files.
- Implement basic programming structures, including variables, functions, conditional statements, and loops.
- Update your website dynamically using event-driven interactions.
- Verify your code is functioning correctly by using browser alerts and the Chrome console

Part 1: Directory Structure and File Setup

To begin Increment 3, you must have successfully completed Increment 2. You should also have a completed design guide and low-fidelity wireframes that demonstrate clear attention to the information architecture, which you are now using to develop your high-fidelity wireframes.

1. Inside your `67250_technologyproject_phaseb` folder, create a new folder named:

```
None
andrewID-increment3
```

2. Copy the contents of your entire increment 2 folder into this new folder
3. Confirm you have the following structure:

```
andrewID-increment3/
│
├── index.html
├── views/
│   ├── checkout.html
│   ├── exhibitions.html
│   ├── explore.html
│   └── etc...
│
├── static/
│   └── (images, etc.)
│   └── style.css
```

4. Inside the `static` folder, create a new file named `script.js`
5. Link `script.js` in the `<head>` of index.html (root folder) including the `defer` attribute that ensures the script runs *after* the HTML loads:

```html
<script src="./static/script.js" defer></script>
```

6. Link the JavaScript file in the `<head>` of every HTML file in the views folder:

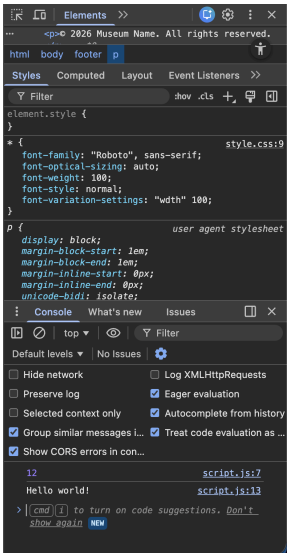
```html
<script src="../static/script.js" defer></script>
```

Part 2: JavaScript Basics (Console Practice)

Variable Declarations and Console Output

1. Open `static/script.js` in VS Code and write the code needed to meet the requirements
2. Add a numeric addition:
    - Declare variables `var x = 5` and `var y = 7`

    - Create `var z = x + y`
    - Print `z` to the console using `console.log()`
3. Add a string concatenation:
    - Create string variables `A = "Hello "` and `B = "world!"`
    - Create `var C = A + B`
    - Print `C` to the console log, using `console.log()`
4. Verify your code is working correctly:
    - Open Chrome → View → Developer → JavaScript Console
    - Console should show:
        - `12`
        - `Hello world!`
    - Example:



Create a Basic Function

5. Define a function using VS code shortcut:

```
JavaScript
function name(params) {

}
```

6. Define a function named `sumnPrint` that accepts two parameters, `x1` and `x2`, as in `sumnPrint(x1, x2)`. The body of the function should perform the following actions:
    - Combine `x1` and `x2` (either by summing them if numeric or concatenating them if strings)

    - Output the resulting value to the console
7. Call it twice by typing:
    - `SumnPrint(x, y);`
    - `SumnPrint(A, B);`
8. Verify your code is working correctly. Console should show:
    - `12`
    - `Hello world!`
    - `12`
    - `Hello world!`

Add a Conditional Statement

9. Write an if/else statement that compares `C.length` to `z`
10. Use the VS code shortcut for ifelse (type the word if and select ifelse from the drop down menu in the frame):

```javascript
if (condition) {
    // Nested condition check
    if (condition) {
        // Code block for nested true
    }
    // End of nested check block
} else {
    // Code block for initial condition false
}
```

11. In the first condition brackets write the first conditional statement that checks if **`C.length > z`**
12. Followed by the rule statement in the curly brackets:
    - If this is true, log **`C`** to the console (if **`C.length`** is greater than **`z`**)
13. Then in the second condition brackets write the second conditional statement that checks if **`C.length < z`**
    - If this condition is true, log **`z`** to the console (if **`C.length`** is smaller than **`z`**)
14. In the last curly brackets state the else rule that applies in case none of the conditions apply.
    - Print `"good job!"` if they are equal
15. Verify your code is working correctly. The console should show `"good job!"`.

Arrays + Loops (Alerts)

1. Create two arrays:

```javascript
L1 = ["Watermelon","Pineapple","Pear","Banana"];
L2 = ["Apple","Banana","Kiwi","Orange"];
```

2. Write a function `findTheBanana()` that
   - Loops through the array (using a for loop)
   - Uses `alert()` when `"Banana"` is found (using an if statement)
3. Call the function with both arrays. More on: **forEach** JavaScript method.
4. Verify your code is working correctly. You should see only one alert for each array.
5. Now rewrite the function using `forEach()` and confirm it works again.
6. Once confirmed, feel free to comment these functions out.

Part 3: Time-Based Greeting (DOM Manipulation)

Add an ID to the Home Page

1. Now you will add a dynamic time based greeting to your home page. In **index.html**, add an `<h2>` element with an id=greeting somewhere near the top of your page *OR* edit the `<h2>` element you may already have:

```html
<h2 id="greeting">Welcome to the MonoMuse Museum</h2>
```

Define the Function greeting()

2. In `script.js`, declare a new variable called now, and assign to it a new time object (by using new Date())
3. Declare a new variable called hour and assign to it the extracted hour form now object by using the method .getHour
4. Define a function `greeting(x)` that:
   - Receives the hour as parameter,
   - Changes the inner html of `<h2> #greeting` based on the following condition:
     - Good morning (hour < 12)
     - Good afternoon (12 ≤ hour < 18)
     - Good evening (18 ≤ hour < 20)

    - Good night (20 ≤ hour < 24 OR hour < 5)

  Important: Implement your conditions in this order to avoid overlapping cases:

    - Good night (h < 5 OR h >= 20)
    - else Good morning (h < 12)
    - else Good afternoon (h < 18)
    - else Good evening
5. Verify your code is working correctly. Refresh `index.html` and confirm the greeting appears with the correct timing.

Fix the Error on the Other Pages (Required Debugging Task)

6. The greeting DOM code may create errors on pages that do not contain the `id="greeting"`. That is because our script.js is running on all pages.
7. Fix this by making sure your greeting code only runs when the element exists (or only on index.html).
    a. Hint for the first fix: Before updating an element, check that it exists on the page, for example, by confirming document.getElementById(...) returns a value.
    b. Hint for the second fix: You can use window.location to create this fix. (See https://www.w3schools.com/js/js_window_location.asp for more information).
8. Verify your code is working correctly:
    - Greeting shows on index.html
    - No console errors on any other pages

Part 4: Dynamic Footer Year(All Pages)

Update Footers

1. On every HTML page, add a paragraph inside the `<footer>`:

```html
<p id="copyYear"></p>
```

2. Remove the year from your footer text so the year is generated by JavaScript

Write addYear()

3. In `script.js`, write a function `addYear()` that:
    - Uses `new Date().getFullYear()`
    - Updates the `copyYear` paragraph

Call addYear() When the Page Loads

4. In the `<body>` tag of every html page, add:

```HTML
onload="addYear()"
```

5. Verify your code is working correctly:
    a. The current year appears in the footer on all pages

Part 5: Browser Check, Console Check and Validation

- Double check that your JavaScript file is linked correctly on all HTML pages
- Confirm the greeting update on the home page and does not create console errors for the other pages
- The footer year is dynamic and works correctly on all pages
- There are no additional console errors
- Check your HTML code by using W3C Validator: https://validator.w3.org/
- Fix major errors

Part 6: GitHub and Deployment

Add Increment 3 to GitHub

1. Select your github repository named 67250_techproject_phaseb
2. In the blue section you see on your screen click “upload an existing file” and drag and drop your andrewID-increment3 folder. Drag the full folder so it is in the root of your repo.
3. After all your files upload, click Commit changes (green button at bottom).

Optional: VS Code will also integrate with GitHub if you’d like to use those features instead.

Deploy on GitHub Pages

1. In GitHub, go to Repository → Settings → Pages
2. Select:
    - Branch: main
    - Folder: / (root)
3. Click Save

Your site will deploy from the root of your repository.

Note: Since Increment 3 lives inside andrewID-increment3, you may need to manually navigate to that folder in the deployed URL. For example, the URL will be something like: https://jeriaq.github.io/67-250_techproject_phaseb/andrewid-increment3/index.html

Important Notes

- You can create a landing page (index.html) that sits in your root
- You do NOT need to create a new repository for each increment.
- You do NOT need to create new branches for each increment.

Part 7: Submission

You do not need to upload anything to Canvas.

To receive credit:

- Your GitHub repository must be public
- The andrewID-increment3 folder must exist
- Files must be properly structured and accessible

TAs will review your GitHub repository directly.

🎉 Congratulations on completing Web Development Increment 3! 👏